Chatting with Building Codes

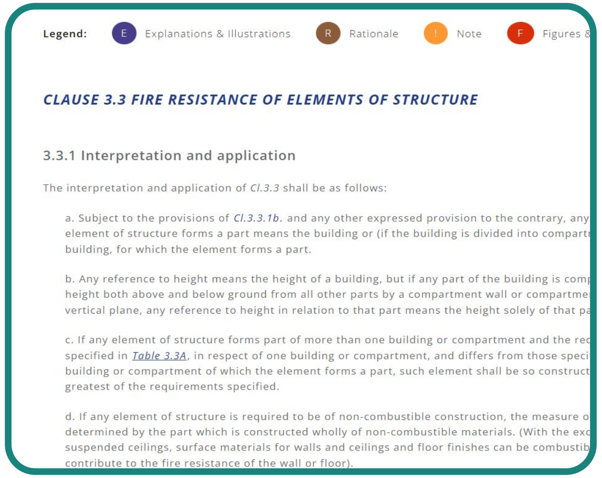
Team 6
Anders Ang
Wei Sheng
Yiing Huey
Gabriel Chek

SINGAPORE ARCHIFEST

AI Hackathon 2024

2024 Singapore Archifest 2024

Problem Statement & Challenges


Legend: E Explanations & Illustrations R Rationale ! Note F Figures &

CLAUSE 3.3 FIRE RESISTANCE OF ELEMENTS OF STRUCTURE

3.3.1 Interpretation and application

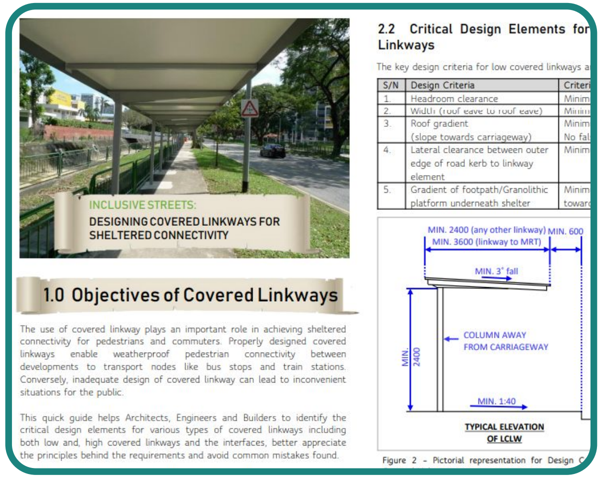
The interpretation and application of *Cl.3.3* shall be as follows:

a. Subject to the provisions of *Cl.3.3.1b*. and any other expressed provision to the contrary, any element of structure forms a part means the building or (if the building is divided into compart building, for which the element forms a part.

b. Any reference to height means the height of a building, but if any part of the building is comp height both above and below ground from all other parts by a compartment wall or compartme vertical plane, any reference to height in relation to that part means the height solely of that pa

c. If any element of structure forms part of more than one building or compartment and the req specified in *Table 3.3A*, in respect of one building or compartment, and differs from those speci building or compartment of which the element forms a part, such element shall be so construct greatest of the requirements specified.

d. If any element of structure is required to be of non-combustible construction, the measure o determined by the part which is constructed wholly of non-combustible materials. (With the exc suspended ceilings, surface materials for walls and ceilings and floor finishes can be combustib contribute to the fire resistance of the wall or floor).



1.0 Objectives of Covered Linkways

The use of covered linkway plays an important role in achieving sheltered connectivity for pedestrians and commuters. Properly designed covered linkways enable weatherproof pedestrian connectivity between developments to transport nodes like bus stops and train stations. Conversely, inadequate design of covered linkway can lead to inconvenient situations for the public.

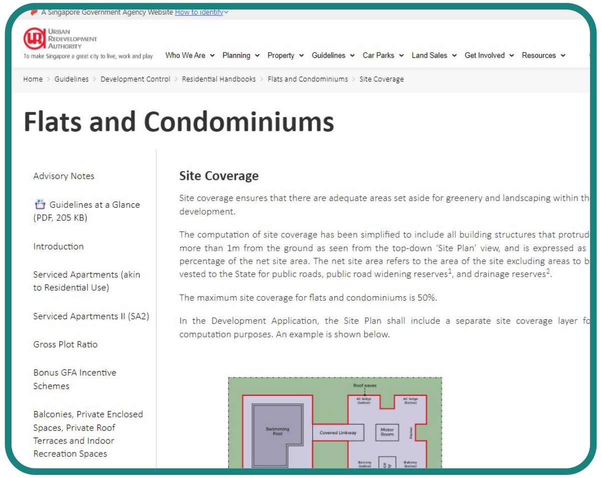
This quick guide helps Architects, Engineers and Builders to identify the critical design elements for various types of covered linkways including both low and, high covered linkways and the interfaces, better appreciate the principles behind the requirements and avoid common mistakes found.

2.2 Critical Design Elements for Linkways

The key design criteria for low covered linkways a

S/N	Design Criteria	Criteri
1.	Headroom clearance	Minim
2.	Width (roof eave to roof eave)	Minim
3.	Roof gradient (slope towards carriageway)	Minim No fal
4.	Lateral clearance between outer edge of road kerb to linkway element	Minim
5.	Gradient of footpath/Granolithic platform underneath shelter	Minim towar

Figure 2 – Pictorial representation for Design C


A Singapore Government Agency Website How to identify

Urban Redevelopment Authority

Who We Are Planning Property Guidelines Car Parks Land Sales Get Involved Resources

Home › Guidelines › Development Control › Residential Handbooks › Flats and Condominiums › Site Coverage

Flats and Condominiums

Advisory Notes

Guidelines at a Glance (PDF, 205 KB)

Introduction

Serviced Apartments (akin to Residential Use)

Serviced Apartments II (SA2)

Gross Plot Ratio

Bonus GFA Incentive Schemes

Balconies, Private Enclosed Spaces, Private Roof Terraces and Indoor Recreation Spaces

Site Coverage

Site coverage ensures that there are adequate areas set aside for greenery and landscaping within th development.

The computation of site coverage has been simplified to include all building structures that protrud more than 1m from the ground as seen from the top-down 'Site Plan' view, and is expressed as percentage of the net site area. The net site area refers to the area of the site excluding areas to b vested to the State for public roads, public road widening reserves[1], and drainage reserves[2].

The maximum site coverage for flats and condominiums is 50%.

In the Development Application, the Site Plan shall include a separate site coverage layer fo computation purposes. An example is shown below.

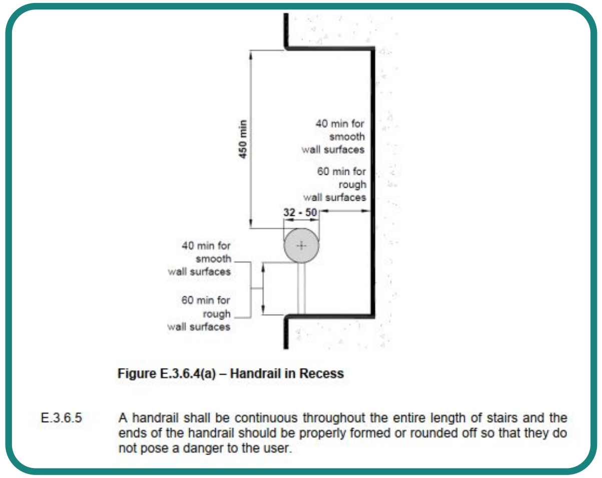


Figure E.3.6.4(a) – Handrail in Recess

E.3.6.5 A handrail shall be continuous throughout the entire length of stairs and the ends of the handrail should be properly formed or rounded off so that they do not pose a danger to the user.

For every architectural project, we must comply with the regulations set out by the relevant authorities.

In Singapore, that means having to refer to multiple PDFs, web pages, formatted in all sorts of different ways by different authorities.

Having to search and process all these building regulations is time-consuming, and sometimes trying to find the appropriate regulations is like finding a needle in a haystack.

Time spent on this "search" activity takes away time from other architectural activities that generate more value for clients and the community.

Problem Statement & Challenges

The problem we have is a problem of search - there's just a lot of data to process and collate from various areas.

And this problem happens to be one that LLMs can solve - so we built a chatbot that allows us to "chat" with building codes.

Objectives & Impact

1. Learn about recent developments in Large Language Models (LLMs) and Information Retrieval (IR) and their potential roles in the architecture discipline.

2. Develop a chat agent that helps quickly retrieve and summarise building code requirements with natural language query by leveraging on what we have learnt.

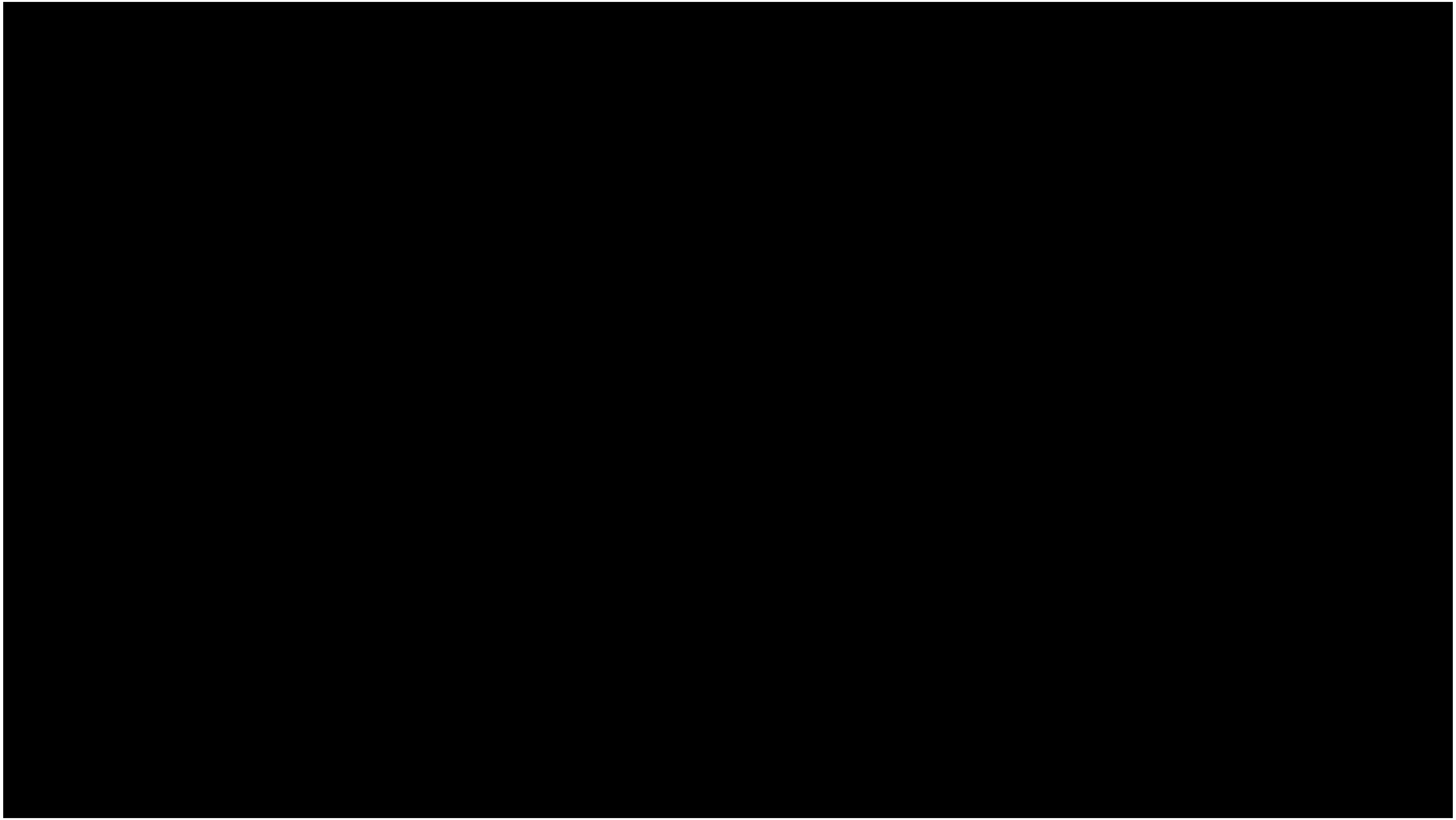
3. Explore various parameters and actions to improve the accuracy of LLMs for Information Retrieval, by:

    1. Exploring different data formats

    2. Exploring different data indexing strategies

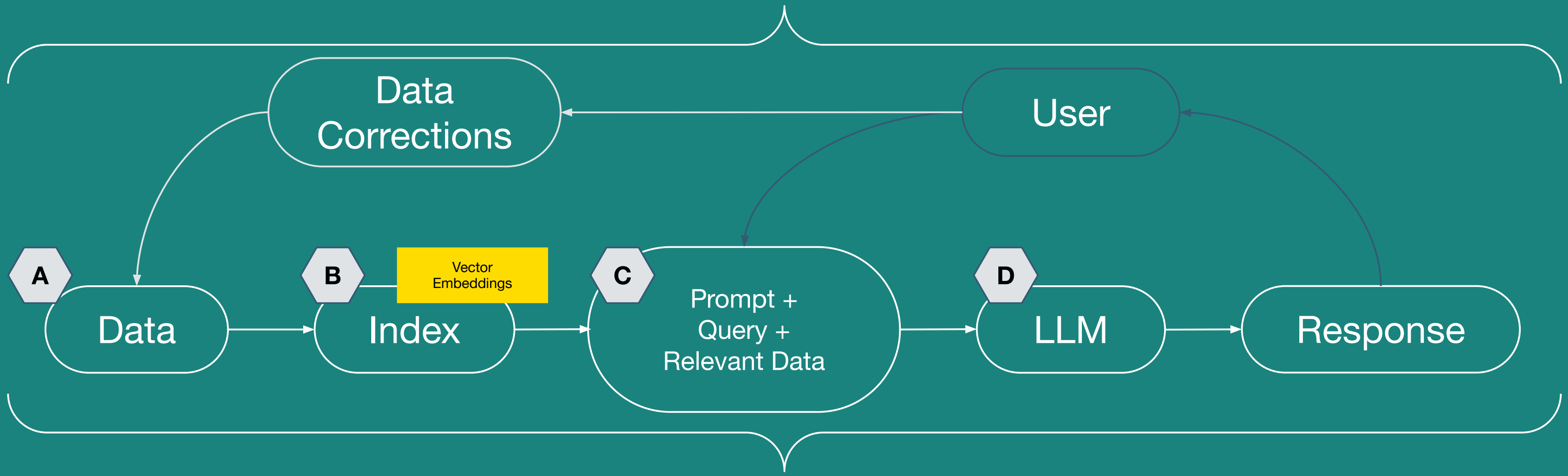
    3. Exploring prompt engineering

Solutions and Tools



Solutions and Tools

Retrieval Augmented Generation (RAG)[1]



Often faster and cheaper (to build), more accurate
More suitable for accurate building code recall, for constantly evolving building code

1. Patrick Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" (arXiv, April 12, 2021), http://arxiv.org/abs/2005.11401.

Strategies

LlamaIndex Framework[2]

A — Data
- Data Formatting
- Data Corrections
- Data Splitting

B — Index
- Embedding Vector model selection
- Embedding Dimensions

C — Prompt Engineering[8]
- Zero-shot Chain of Thought[3 4]
- Generated Knowledge[5]
- Self -consistency[6]

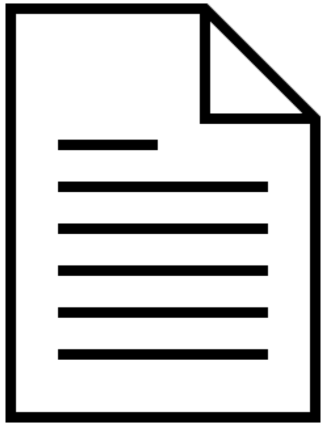
D — LLM
- LLM Selection
- Chat Engine Configuration[7]
- more…

Data framework for LLM-based applications.
Not restricted to any provider.

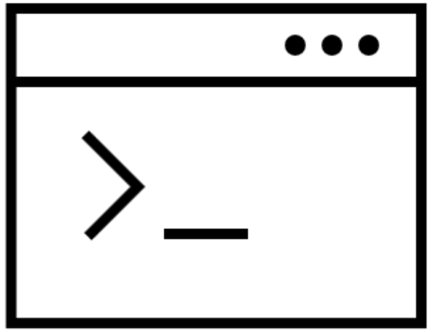
Limitations

Data needs to be represented in a computer-readable format to be parsed and retrieved accurately.

Building codes from various agencies are presented in vastly different structures and formats, that may or may not allow for accurate retrieval and parsing in whole by computers.



A prominent example of this is the ubiquitous pdf format - while a great format for portable, human-readable documents, when parsed by a computer they often present broken-up sentences and text that is out of order. This often results in the retrieval of wrong information by the LLM.



Limitations

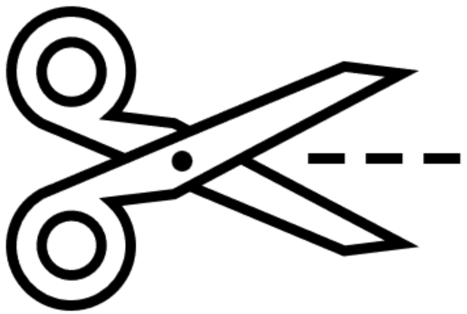
Data that conforms to commonly used standards (eg. if dates are formatted to ISO 8601) are less likely to be retrieved and interpreted incorrectly

LLMs that we tested very often misinterpreted the BCA Approved Doc’s versioning - version 7.05 in BCA’s pdf was interpreted as version 7.0. Common versioning standards would have omitted the “0” and wrote it as version 7.5, or version 7.0.5.

Limitations

Data should be split up into relevant “chunks”. In our tests, full PDF documents often perform worse than PDFs separated into individual clauses.

Indexing “chops up” a document into smaller parts (nodes) based on “chunk” sizes .This may break up parts of the building code that are related to each other - eg breaking up a clause mid-sentence. As these nodes are all under the same document, the relationship between nodes is diluted.
Instead, we separated documents into their individual clauses. These docs still get “chopped up”, but these nodes are then tied to the same metadata (eg. all clauses under E.1 go under the E.1 document). This improves our responses.



Limitations

YAL BuildSG

ORIGINAL PDF

A GENERAL

A.1 INTRODUCTION

A.1.1 The framework for performance-based building code is set out in the Building Control Regulations 2003 (referred to in this Document as the Regulations). The Fifth Schedule of the Regulations sets out the objectives and performance requirements that must be complied with in the design and construction of building works (referred to in this Document as "prescribed objectives and performance requirements"). The objectives set out community expectations of a safe, accessible and energy efficient building. The performance requirements outline the level of performance, which must be met in order for a building to meet the objectives.

A.1.2 This Approved Document provides a set of 'acceptable solutions' that meet the prescribed objectives and performance requirements. The prescribed objectives and performance requirements are deemed to be satisfied if the design and construction of a building comply with the acceptable solutions.

A.1.3 Alternatively, a person may utilise alternative solutions in respect of the design and construction of any building if these solutions satisfy the prescribed objectives and performance requirements. Alternative solutions are solutions that entail the use of any design, material or construction method that differs completely or partially from those in the acceptable solutions.

A.2 ABBREVIATIONS AND SYMBOLS

A.2.1 The following abbreviations and symbols are used in this Document –

Abbreviation or Symbol	Definition
BS	British Standard
CP	Code of Practice
°K	degree Kelvin
kW	kiloWatt
kg	kilogram
m	metre
m^2	square metre
mm	millimetre
SS	Singapore Standard

Building and Construction Authority
Approved Document – Acceptable Solutions V7.05 Page 1 of 84

SEGMENTED PDF

A GENERAL

A.1 INTRODUCTION

A.1.1 The framework for performance-based building code is set out in the Building Control Regulations 2003 (referred to in this Document as the Regulations). The Fifth Schedule of the Regulations sets out the objectives and performance requirements that must be complied with in the design and construction of building works (referred to in this Document as "prescribed objectives and performance requirements"). The objectives set out community expectations of a safe, accessible and energy efficient building. The performance requirements outline the level of performance, which must be met in order for a building to meet the objectives.

A.1.2 This Approved Document provides a set of 'acceptable solutions' that meet the prescribed objectives and performance requirements. The prescribed objectives and performance requirements are deemed to be satisfied if the design and construction of a building comply with the acceptable solutions.

A.1.3 Alternatively, a person may utilise alternative solutions in respect of the design and construction of any building if these solutions satisfy the prescribed objectives and performance requirements. Alternative solutions are solutions that entail the use of any design, material or construction method that differs completely or partially from those in the acceptable solutions.

A.2 ABBREVIATIONS AND SYMBOLS

A.2.1 The following abbreviations and symbols are used in this Document –

Abbreviation or Symbol	Definition
BS	British Standard
CP	Code of Practice
°K	degree Kelvin
kW	kiloWatt
kg	kilogram
m	metre
m^2	square metre
mm	millimetre
SS	Singapore Standard

Limitations

Data and queries should be as explicit as possible.

LLM finds patterns in data and attempts to construct a response. It may find the wrong patterns - if the same word is used in different contexts, it may retrieve data from the wrong context.

In our tests, when asking about recessed voids using Gemini 1.0, it also gives information regarding recesses for handrails, which is irrelevant to the query.

Query formulation can be improved, or, building code can be made more explicit, with differentiated terminology, to solve these issues.

What Next

Advancements can come in the following forms:

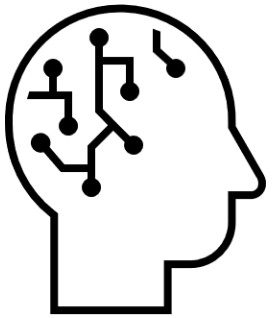
1. Quick reference Chatbot that can be continuously refined through community feedback.

2. Better formatting and standardisation of future code releases.

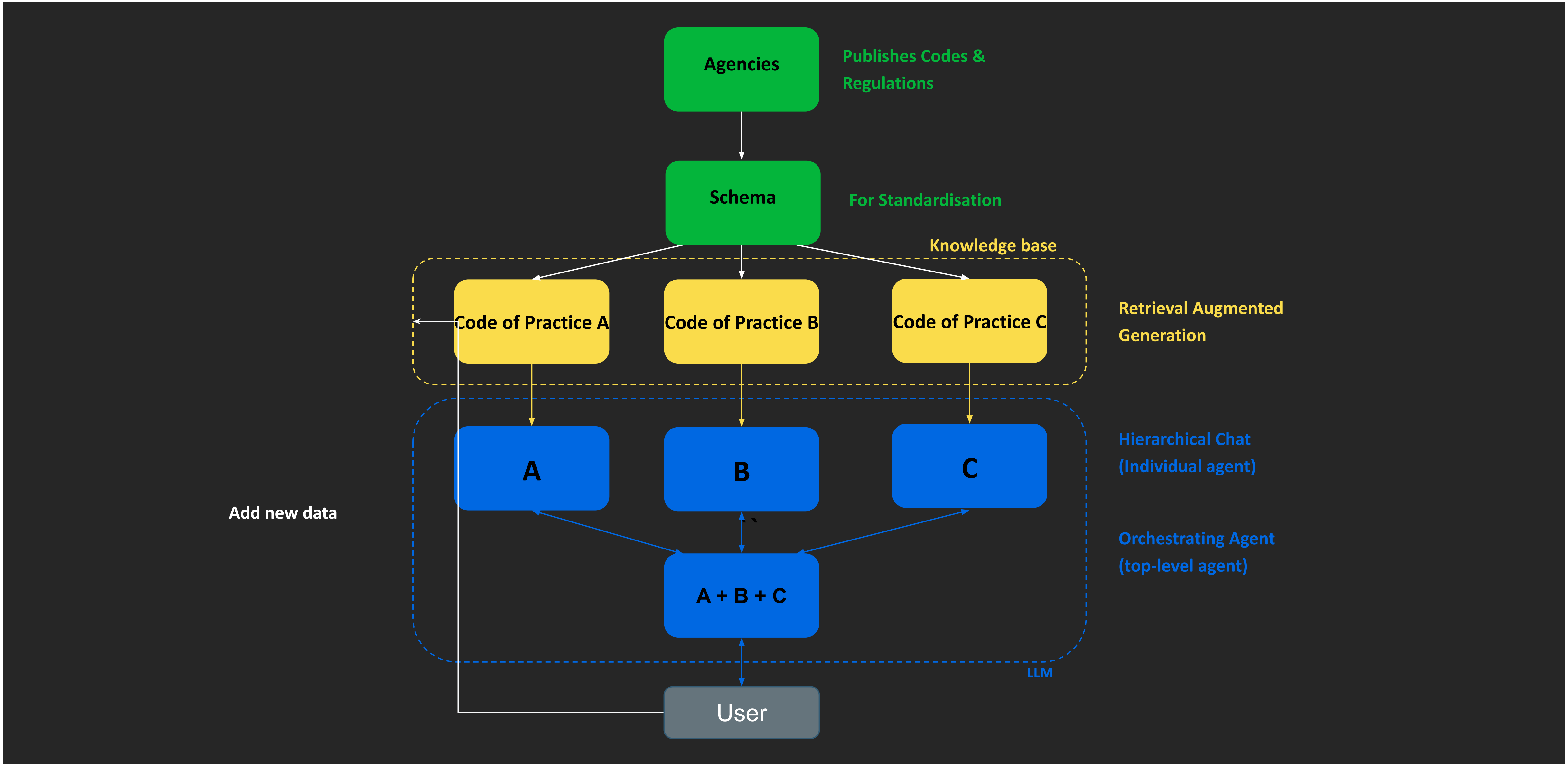
3. Developing a Chatbot that can help convert future code releases to the proper format.

4. Developing a Chatbot in which through providing parameters of the project, generate a checklist of code requirements specific to project parameters.

5. Developing a Chatbot that can interpret the code for special situations by referencing past case studies and examples from overseas.



Example of an Agentic system



SINGAPORE ARCHIFEST

Thank you

2024 Singapore Archifest 2024